

My experience with CUDA brought expected challenges. After a brief discussion with Dr. Pelosi in Fall '25, I attempted to explore CUDA on my own. My exploration ended abruptly after continued issues trying to install CUDA properly. Microsoft Copilot suggested that **only** Visual Studio could handle CUDA, but then any current version of Visual Studio was too updated to support CUDA. So I would have to backlog my Visual Studio version to handle CUDA, which would never work. I experience this same issue during this assignment, but eventually changed some environment settings to update my terminal's default configuration to support the correct version of CUDA. Not only does this allow VSCode to support CUDA, but it actually **prevents** Visual Studio from accessing it. So, after three (3) hours of setting up the CUDA environment, I had success. As stated in Canvas, that was the hard part. The actual design and implementation of this project was shockingly easy. I expected some sort of trivial parallel processing to fully utilize the capabilities of CUDA, but we simply used CUDA libraries to read device properties. CUDA methods have incredible documentation, so return types and use-cases were easy to find, leading to a straight-forward implementation. Besides a hiccup with my version of CUDA not supporting any type of clock rate, this assignment was actually a first-attempt success. In the future, I hope to have a project in CUDA where I am forced to learn and utilize parallel-processing. Through the "sc11-cuda-c-basics.pdf" on Canvas, I attempted to learn the block and thread properties of parallel computing, but was left relatively confused by it. I think, with a proper project and forced learning, I could absorb this knowledge.